

Fellowship Plan

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As a PhD student excited about research and teaching, my longer-term goal is to be a professor in computer science. To advance this objective, the following research skills and experiences, which are my shortcomings, are essential: mentoring junior students, teaching, collaborating with more colleagues, authoring a research proposal, and improving my presentation skills. In this statement, I will explain why I believe professor Nadia Polikarpova will be a good mentor for me and how we plan to work on the mentioned research skills and experiences during the fellowship.

Compatibility of research interests. My research aims to propel program synthesizers—automated tools generating programs from specification—to real world scenarios by exploring effective specification mechanism for users to express their intents, and studying when and why synthesizers fail on given problems. At the same time, program synthesis is also one of the research interests of Nadia. I am familiar with Nadia’s synthesis-related work, and one of my prior work was inspired by *Resource-Guided Program Synthesis*, Nadia’s paper appeared in PLDI’19. So advices from Nadia would be very helpful for me and we can work together on research topics that align with our interests.

Orientation. Mutual expectations will be discussed and agreed prior to starting the proposed research: 1) the amount of independence the postdoctoral researcher requires, 2) interaction with collaborators and their graduate student research assistants, 3) scheduling weekly group meetings to discuss research progress, and 4) productivity including the importance of scientific publications.

Career Counseling will be provided to help the postdoctoral researcher in setting and achieving his career goals beyond his postdoctoral training. Realistic career options for the postdoctoral researcher will be discussed, including tenure- track positions and research opportunities in industry, as well as how to best prepare for these positions. In addition to guidance provided by the mentors, the postdoctoral researcher will be encouraged to discuss career options with other researchers at UCSD.

Mentoring and Teaching Skills will be developed in the context of regular meetings within UCSD research groups. There will be opportunities to work with junior graduate students and assisting them with their research projects. The mentor will provide guidelines about how to motivate junior students and give them proper advices. In addition, I plan to give lectures about program synthesis in the mentor’s class and expect the mentor to share experience of teaching and preparing course material. Then the mentor will provide feedback about my teaching.

Opportunities of Collaborating with Colleagues. There are many excellent professors and students working on programming languages and formal methods in UCSD. I expect that we will describe our works to each others within the research groups and assist each other with solutions to challenging research problems. With interesting solutions raised, we can collaborate on a new research project. Besides the PL group, there will also be opportunity to listen to other group’s seminar and talk to researches in other directions which may lead to research project about applying program synthesis in other CS directions.

Experience with Preparation of Grant Proposals will be gained by direct involvement in proposals prepared by the mentor. The postdoctoral researcher will have an opportunity to learn best practices in proposal preparation including identification of key research questions, definition of objectives, description of approach and rationale, and construction of a work plan, time-line, and budget. Also, I expect that I will receive training in other professional activities such as how to review a paper, how to prepare a tutorial and how to organize a seminar.

Presentations Skills are expected to result from the work supported by the fellowship. The mentor will provide advices and feedbacks on paper writing and tool implementing. I expect to receive guidance and training in the preparation of manuscripts for scientific conferences such as POPL, CAV, and PLDI. There will also be opportunity to give practice talk in the Programming System group and listen to the feedbacks which can help me to improve my presentation and communication skills.

Success of the Mentoring Plan will be assessed by monitoring the personal progress of the postdoctoral researcher by tracking the progress toward his career goals.